

GraphHoldout: publication-time gene-disease ranking with cold-start refusal

Candidate-specific working report, September 25, 2026. This is separate from the aggregate item-27 manuscript and is not a validated gene-discovery paper. The build produces a reproducible ranker, source inventory and leakage audit. It does not prove causal disease interactions or use physical protein-interaction evidence.

Abstract

Graph methods can look accurate if a target gene-disease edge is duplicated in reverse, if evidence from the same paper crosses train and test, or if future curated labels are treated as negative until their publication. We used the original CTD curated gene-disease report and PubMed publication-year metadata to construct 556 evidence rows for four named diseases. A 2012 cutoff produces 356 unique training pairs and 114 novel later pairs. Of the later pairs, 106 involve genes absent from the training graph, so a ranker limited to known nodes cannot retrieve them. Among eight reachable later positive pairs, top-10 recall is asthma 2/2, breast neoplasms 0/5, prostate cancer 0/1, and diabetes unevaluable. Unknown pairs were never treated as experimental negatives; no precision or AUROC is claimed. A current Open Targets query provides additional context but not an independent historical validation, since underlying literature and CTD contributions may overlap. The result is a working temporal-leakage/cold-start audit, not a validated biological predictor.

1. Question and task boundary

The original candidate called for graph-based gene-disease ranking with source- and time-disjoint edge holdouts. Named biological interaction evidence must remain distinct from text-mined co-occurrence. A strict version would train on a genuinely historical graph snapshot from one source and assess prospectively new, independently curated edges from another source, with interaction provenance. This build tests only a narrower question: under a publication-year split *within one modern CTD snapshot*, can a simple training-only neighbor ranker recover subsequently published CTD-curated gene-disease associations for genes already in the training graph?

This question matters because a model evaluated on random edges can receive nearly the same biological relation through reverse arcs or duplicated publications. Even when leakage is removed, a cold-start gene cannot be ranked by a graph-only method that has no representation for it. The implemented software makes those limitations observable rather than returning a seductive performance score on mislabeled unknowns.

2. Data lineage and exact identifiers

The direct source is the [CTD curated gene-disease report](#), downloaded as a compressed TSV. `data/manifest.json` records its SHA-256 digest. We read only rows with a direct curated evidence code and PMID, not inferred or text-mined edges. Four disease IDs were fixed by exact MESH term: D001249 asthma, D001943 breast neoplasms, D003920 diabetes mellitus and D011471 prostate neoplasms. We chose every third numerically sorted PubMed ID among matching rows, capped at 275 distinct PMIDs, before querying dates. This is a tractability sample, not a representative random biological sample. It produces 556 gene-MESH-PMID evidence rows from 275 publications. The row-level `data/edges.json` records gene symbol, NCBI Gene ID, disease ID, evidence type, PMID and publication year.

Publication years were read through the [NCBI PubMed E-utilities summary endpoint](#). The year of an article is not necessarily when CTD added a curated edge. This distinction is fundamental: a 2026 CTD snapshot cannot be made into a genuine 2012 graph merely by filtering contemporary rows on the original article's date. The experiment is a retrospective illustration of a split, not a historical replay. Rows with no usable year are excluded rather than assigned one.

3. Edge representation and holdout

Each gene-disease relation is represented once by canonical (NCBI Gene ID, MESH ID). A publication through 2012 puts that relation in training. A later publication contributes a novel test pair only when that exact relation was absent from training. We also require the source PMID sets on opposite sides to be disjoint, rejecting any overlap. Representing one canonical relation makes a reverse disease-gene arc equivalent to the same edge; it cannot be left in training when the target pair is held out. An explicit test covers this recurrence and a second test rejects a PMID shared across split years.

The candidate universe for each disease includes only genes observed in training through 2012 whose target disease edge is not already known. All other candidate pairs remain **unlabeled**, not measured biological negatives. The ranker can put a later positive high in this universe, but top-10 recall is not precision, because the noncurated remainder can contain true associations. We separately count later positive pairs whose gene never appeared in training as cold-start failures rather than hiding them from the denominator of task feasibility.

4. Model and metrics

For each candidate gene and disease, the score sums the size of shared training-gene neighborhoods between the target disease and every other training disease to which the gene connects, plus 0.01 times the candidate gene's training disease degree. This is a simple two-hop graph heuristic, not a graph neural network. The 0.01 degree term and top-10 cutoff were fixed as software constants; the resulting values have not been optimized against the held-out pairs. The exact ranking and tie-breaker by gene ID are reproducible from `evaluate.py`.

For a target disease with reachable later positives, positive recall@10 equals the number of those positives ranked in the first ten candidate genes divided by the number of reachable later positives. The report also gives the expected top-10 fraction under a uniform random permutation of the same candidate list. That random expectation is a comparison of ranks, not an empirically validated negative control or a statistical significance test. Where a disease has zero reachable positives the recall is explicitly null.

5. Primary outcomes

Disease	Reachable novel positives	Candidate genes	Positive recall@10	Random expected recall@10
Asthma	2	306	2/2	0.0327
Breast neoplasms	5	164	0/5	0.0610
Diabetes mellitus	0	323	Unassessable	0.0310
Prostate neoplasms	1	183	0/1	0.0546

The graph contains 356 unique pre-cutoff pairs and 114 novel post-cutoff pairs. Only eight test positives involve genes available to rank. A high-looking 2/2 for asthma is too small and selected to support generalization, especially against six other reachable positives with zero hits. The **106 cold-start pairs** make a stronger practical point: a graph-only ranker limited to known genes has no defined prediction for most subsequently curated edges in this sample. No gene is nominated for laboratory testing from this exploratory output.

6. External source context without false independence

The [Open Targets Platform GraphQL endpoint](#) was queried for four current MONDO concepts, mapped to the four CTD MESH terms by concept name. Exact-symbol overlaps between sampled CTD genes and the platform's current top-100 associated targets were 9/39 asthma, 19/220 breast neoplasm, 27/197 prostate cancer and 4/14 diabetes mellitus. `data/external_source.json` retains the query, disease ID, returned ranking, response digest and matched symbols.

That endpoint is not an independent 2012 validation. It may include the same papers or CTD-derived evidence, and symbols are not stable accession mappings. Current target ranks include evidence accumulated after 2012. A second website with overlapping source content is not a second scientific discovery. This analysis therefore checks modern cross-service context only and is not added to the temporal benchmark denominator.

7. Missing interaction and source gates

The CTD rows are curated associations such as marker/mechanism or therapeutic links. They are not direct, named physical gene-product interactions. A path through CTD disease neighbors can suggest co-annotation; it cannot establish protein binding, an intervention effect or causal disease mechanism. A future mechanistic graph would need interaction records with their experiment type, partners, provenance and date, and it would keep text-mined co-occurrence in a separate layer. No such interaction layer was added here, so the build's graph score is expressly a disease co-annotation heuristic.

A genuinely source-disjoint evaluation would require a historical training snapshot that can be dated by curation time and an independently generated future test set with evidence overlap checked at paper, gene and disease level. At least 20 reachable positives per disease and several independent diseases would allow a less fragile recall estimate. We have

no historical CTD curation snapshot or independently adjudicated disease-gene test set, and not enough reachable positives in this sample. These are named missing gates, not promises of later success.

8. Reproducibility and adversarial checks

Run `python3 projects/graph_holdout/prepare.py` to download CTD and query PubMed dates; compare CTD source hash to committed `data/manifest.json`. Run `python3 projects/graph_holdout/evaluate.py` on committed `data/edges.json` to regenerate rankings and primary result JSON. Run `python3 projects/graph_holdout/external_source.py` to query current Open Targets ranks, noting those results can change. Run `python3 -m pytest -q projects/graph_holdout` for tests of reverse-relation leakage, shared-PMID rejection, unseen-gene handling and the cross-source query shape. CTD's publication/citation and linking terms need separate legal review for downstream redistribution; see <https://ctdbase.org/about/legal.jsp>. The repo retains a small sampled derivative, not the full CTD report.

9. Interpretation

The task is not solved. The usable result is a transparent auditor that refuses to turn absent labels into negatives, keeps one edge and its reverse together, separates publication sources across time, and exposes the cold-start denominator. The apparent asthma top-10 result cannot offset the breast/prostate failures, tiny denominators or single-source retrospective leakage risk. No clinical, mechanistic or therapeutic inference is supported. A new prospectively registered historical-source and experimentally grounded graph would be a different study, not a retrofit of this result.

Sources and files

Source files are the [CTD curated association download](#), [PubMed E-utilities](#), and [Open Targets GraphQL API](#). Exact source digests, used rows and result counts are in `data/manifest.json`, `data/edges.json`, `data/results.json`, and `data/external_source.json`. Scientific source count is at most three, far below the user's per-project 40-tool bar; model libraries and access infrastructure are not counted as extra biological services.

Post-report PMID provenance audit, September 25

The separate `verify_pmids_metadata.py` audit individually retrieved 120 PubMed ESummary records, the first 60 numeric IDs on each side of the 2012 publication-time split. All 120 IDs, response-byte SHA-256 digests, parsed publication dates and linked row counts are in `data/pmids_metadata_audit.json`; every year matches the existing CTD edge table. This is a check against batch parsing errors and bibliographic accession identity. It is **not** 120 independently fetched biological association datasets and does not turn a modern CTD snapshot into a historical one. The original ranker results and refusal stand.

Actual March 2016 GWAS archive eligibility

The [GWAS Catalog original March 6, 2016 release](#) is a dated source, unlike the current CTD snapshot retrospectively split by paper year. Its original 24,413 SNP-trait rows were pinned by SHA-256. Exact trait names, not synonym searches, yield 55 Asthma, 199 Breast cancer, 201 Prostate cancer, 97 Type 1 diabetes and 340 Type 2 diabetes rows. Only 31, 146, 104, 60 and 205 rows respectively carry a single unambiguous numeric SNP-locus Entrez mapping; the rest have missing or multi-ID mappings. Three original CTD breast cancer publication IDs and one prostate cancer publication ID recur in the corresponding GWAS trait subset. Asthma and the separate Type 1/2 diabetes subsets show no same-PMID match within our small CTD panel, but that alone does not establish independent evidence, trait equivalence or temporal train/test integrity. A GWAS SNP's locus-mapped gene is not a directly assayed causal gene-disease link, and a plain-text trait is not a verified MeSH disease crosswalk. In particular, the CTD target's broad Diabetes Mellitus term cannot silently absorb both Type 1 and Type 2 diabetes. We do not treat these rows as biological association negatives or as source-disjoint positive truth, and calculate no new recall or AUROC. The source digest, exact rules, per-target publication overlap, executable scope audit and tests are in `data/gwas_archive/` and `gwas_archive_scope.py`. This is a substantive negative validation finding, not a newly validated predictor.